Interdiction Models for Disrupting Poached Tiger Part transportation Networks

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1. Introduction

Tiger poaching is a serious wildlife trafficking problem that threatens the survival of this iconic species. After these magnificent creatures fall victim to poaching, various body parts are illicitly smuggled to different parts of the world, fueling a lucrative trade that pushes tigers closer to the edge of extinction. While significant efforts have been made to combat this issue through international network interdiction, there exists a crucial yet often overlooked aspect in our battle against this illicit trade: local networks. Local poaching networks play a vital role in the illegal tiger trade. Poachers work with local communities to transport tiger parts from the forest to urban areas, where they are then smuggled to other parts of the country or internationally. Interdicting local networks is essential to reducing tiger poaching and protecting this species. By developing effective interdiction models, we can help law enforcement agencies to disrupt the illegal tiger trade. To address this issue comprehensively, we propose a class project that will focus on creating an interdiction model aimed at disrupting these local networks.

Our project focuses on developing a versatile shortest path network interdiction model with two distinct applications. In the first scenario, we consider the arc length as the cost, emphasizing the minimization of physical distance traveled by traffickers. In the second scenario, we treat the arc length as the logarithm of the probability of detection, aiming to strategically disrupt routes to increase the total probability of detection. This dual-application model provides a comprehensive tool for addressing different aspects of wildlife trafficking, offering decision-makers flexible solutions based on specific priorities—whether minimizing distance or maximizing the probability of detection. We will gather essential data from a reserve forest area, allowing us to construct a data set that will be instrumental in evaluating the effectiveness of both models in interdicting local networks.

This project presents a comprehensive plan to address the critical issue of tiger poaching in a vulnerable reserve forest area known for its high levels of illegal wildlife trade. We initiated this project with a primary focus on the ethical and meticulous collection of data that forms the backbone of the poached tiger part transportation network. This data will encompass vital information about the transport of tiger parts from the reserve forest to urban areas and other starting points within the international network. To gain a comprehensive understanding of the network, we will closely examine the spatial arrangement of nodes and arcs within a specific region of the forest. This focused attention on the positioning of nodes and arcs in this particular area will contribute to a thorough analysis of the network's characteristics. Key information that we aim to collect includes the routes used for trafficking, and the probability of detection of the trafficker using a particular route.

2. Model formulation

For the network interdiction problem, we will adopt a bi-level optimization approach, focusing on shortest-path scenarios. In the initial formulation, the objective is to maximize the length of the shortest path from source (s) to target (t) in an undirected network. In this model, the trafficker seeks to minimize travel costs, while the interdictor aims to maximize the minimum shortest path.

2.1. ***Model : Maximizing the minimum shortest*** $s-t$ ***path length:***

This model is defined on a un-directed graph $G(N, A)$, where N is the set of nodes and A is the set of arcs. The nominal length of arc $(i, j) \in A$ is denoted as $c_{ij} \geq 0$. The cost is symmetric for any arc, i.e., for any arc $(j, i) \in A$, $c_{ji} = c_{ij}$. We will begin by establishing a directed network. Subsequently, arcs will be duplicated in the reverse direction, ensuring bidirectionality and transforming the network into an undirected structure. Interdiction increases the arc's length to $c_{ij} + d_{ij}$, where $d_{ij} \geq 0$. If interdiction actually destroys arc (i, j), rendering it impassable, the same model applies by setting d_{ij} to a sufficiently

Table 1: Notations for the model

Network and Sets	
$G(N, A)$	Directed network with node set N and arc set A
$FS(i)$	Set of arcs leaving node i
$RS(i)$	Set of arcs entering node i
Parameters	
B	Total resource available for interdiction
r_{ij}	Resource Required for interdicting arc $(i,j)\&(j,i) \in A$
c_{ij}	Nominal integer length of arc $(i,j) \in A$
d_{ij}	Added integer delay if arc $(i,j) \in A$ is interdicted
p_{ij}	Probability of detection if the trafficker traverses arc $(i,j) \in A$ when the arc $(i,j) \in A$ is not interdicted
q_{ij}	Probability of detection if the trafficker traverses arc $(i,j) \in A$ when the arc $(i,j) \in A$ is interdicted
Decision Variables	
x_{ij}	1, if arc $(i,j) \in A$ is interdicted; 0, otherwise
z_{ij}	1, if arc (i,j) & $(j,i) \in A$ both are interdicted; 0, otherwise
y_{ij}	1, if the trafficker traverses $(i,j) \in A$; 0, otherwise

large value. Interdicting a specific arc (i,j) implies that the corresponding arc (j,i) is also interdicted. We consider that only arc interdiction is possible in this problem. The sets, parameters and the decision variables for Model-1 is defined in Table 1.

$$\text{[MXSP-P]}\ z^* = \max_{\mathbf{x}\in X} \min_{\mathbf{y}} \sum_{k\in A} (c_{ij} + x_{ij}d_{ij})y_{ij} \tag{1}$$

$$\text{s.t.} \sum_{(i,j)\in FS(i)} y_{ij} - \sum_{(i,j)\in RS(i)} y_{ji} = \begin{cases} 1 & \text{for } i = s \\ 0 & \forall i \in N - \{s,t\} \\ -1 & \text{for } i = t \end{cases} \tag{2}$$

$$y_{ij} \geq 0 \quad \forall (i,j) \in A, \tag{3}$$

$$
\begin{aligned}
X = \{\mathbf{x} \in \{0,1\}^{|A|},\, \mathbf{z} \in \{0,1\}^{|A|/2} \mid & \\
& \sum_{(i,j)\in A} r_{ij} z_{ij} \leq B, \\
& z_{ij} = x_{ij}, \quad \forall (i,j) \in A, \\
& z_{ij} = x_{ji}, \quad \forall (i,j) \in A\}
\end{aligned} \tag{4}
$$

Equations (1) - (4) represent the initial primal formulation of this problem, referred to as MXSP-P, which is a bi-level formulation. In MXSP-P, the trafficker aims to minimize the length of the shortest path in the network, while the interdictor seeks to maximize the minimum shortest path length for a given pair of source and destination nodes, denoted as $s - t$. The inner objective involves minimization, while the outer objective involves maximization.

To simplify the formulation and transform both the inner and outer problems into an equivalent maximization problems, we replace the inner minimization problem with its dual. We denote this modified formulation as MXSP-D. If we (i) set $\mathbf{x}$ to a fixed value, (ii) derive the dual of the inner minimization in [MXSP-P], (iii) apply a few straightforward modifications, and (iv) subsequently release the fixed value of $\mathbf{x}$, the ensuing MIP yields MXSP-D as shown in equation (5) - (8).

$$
[\text{MXSP} - \text{D}] z^* = \max_{\mathbf{x},\pi} \pi_t - \pi_s \tag{5}
$$

$$
\text{s.t.} \quad \pi_j - \pi_i - d_{ij} x_{ij} \leq c_{ij} \quad \forall (i,j) \in A \tag{6}
$$

$$
\pi_s = 0 \tag{7}
$$

$$
\mathbf{x} \in X \tag{8}
$$

We can assume $\pi_s = 0$ because the inner minimization of MXSP-P includes at least one redundant flow-balance constraint, a common feature in network flow models containing a balance constraint for each node. Additionally, it is worth noting that the

dual variables π are unconstrained in sign. By reversing their indicated signs compared to convention, we can interpret π as the post-interdiction shortest-path length from s to t.

To accommodate multiple sources and sinks in the network interdiction problem, we propose introducing two new nodes, sup_s and sup_t. All sources are connected to sup_s, and all sinks are connected to sup_t. The arcs connecting sources to sup_s and sinks to sup_t have zero cost, and their interdiction resource requirements are set to very large values, effectively making them immune to interdiction.

The node set N is updated to include these new super-source and super-sink nodes, i.e., $N = N_{\text{regular}} \cup \{\text{sup}_s, \text{sup}_t\}$.

The arc sets are also updated:

$$A = A_{\text{regular}} \cup \{(i, \text{sup}_s) \mid i \in S\} \cup \{(\text{sup}_t, j) \mid j \in T\}$$

$A_{\text{regular}} = \text{Set of arcs in the original network excluding the new connections to sup}_s \text{ and sup}_t\}$

The modified objective function and constraints should consider these new arcs, ensuring that the optimization process correctly reflects the network's structure with multiple sources and sinks.

This approach allows for a flexible representation of the problem with enhanced connectivity, facilitating a comprehensive optimization strategy that considers the interplay between the introduced super-nodes and the rest of the network.

2.2. ***Special case: Maximizing the minimum detection probability of the traffickers***

We can leverage the foundational model to address a specific case centered on the detection probability of traffickers. In this specialized instance of the original formulation, the trafficker seeks to minimize the probability of detection while navigating through the network. Simultaneously, the interdictor aims to maximize the minimum probability of

detection for traffickers. This scenario remains a bi-level optimization problem, and with minor adjustments to the cost structure from the initial formulation, we can effectively capture and model this particular situation. The special model is crafted to maximize the probability of detecting the trafficker, taking into account the optimal decisions $\mathbf{y}_{ij}$ made by the trafficker. These decisions are incorporated into the lower-level problem while adhering to a budget constraint. In the trafficker's problem, the objective is to minimize the probability of detection while traversing the network. It's important to note that this model is specifically tailored for an undirected network.

The probability of detection when the trafficker traverses the network is given by:

$$\prod_{(i,j)\in A} (q_{ij}x_{ij} + p_{ij}(1 - x_{ij})).$$

However, this function is inherently non-linear. To introduce linearity, we apply the logarithm to the probabilities, leading to the following linearized function:

$$\sum_{(i,j)\in A} (\log(q_{ij}) \cdot x_{ij} + \log(p_{ij}) \cdot (1 - x_{ij}))\,.$$

This linearized version of the model retains the essence of the original objective while allowing for a linear optimization approach. The modified formulation becomes:

$$\text{[MXMPD-P]} \quad z^* = \max_{\mathbf{x}\in X} \min_{\mathbf{y}} \sum_{(i,j)\in A} (\log(q_{ij}) \cdot x_{ij} + \log(p_{ij}) \cdot (1 - x_{ij})) * y_{ij} \tag{9}$$

$$\text{s.t.} \quad \sum_{(i,j)\in \mathrm{FS}(i)} y_{ij} - \sum_{(i,j)\in \mathrm{RS}(i)} y_{ji} = \begin{cases} 1 & \text{for } i = s \\ 0 & \text{for all } i \in N - \{s,t\} \\ -1 & \text{for } i = t \end{cases} \tag{10}$$

$$y_{ij} \geq 0 \quad \forall (i,j) \in A, \tag{11}$$

$$X_1 = \{\mathbf{x} \in \{0,1\}^{|A|} \mid \sum_{(i,j)\in A} r_{ij}x_{ij} \leq B\} \tag{12}$$

If we compare the structure of this special case with the original formulation, we observe that c_{ij} is equivalent to $\log(p_{ij})$, and $(c_{ij} + x_{ij}d_{ij})$ can be analogously represented as $\log(q_{ij}) \cdot x_{ij}$. This model can also be reformulated using the dual technique discussed for the parent model. The modified problem is expressed as:

$$[\text{MXMPD} - \text{D}]z^* = \max_{\mathbf{x},\pi} \pi_t - \pi_s \tag{13}$$

$$\text{s.t.} \quad \pi_j - \pi_i - \log(q_{ij}) \cdot x_{ij} \leq \log(p_{ij}) \cdot (1 - x_{ij}), \quad \forall (i,j) \in A \tag{14}$$

$$\pi_s = 0 \tag{15}$$

$$\mathbf{x} \in X_1 \tag{16}$$

3. Methods:

3.1. ***Handling bi-level programming***

In this project the actual shortest path network interdiction problem is bi-level. The bi-level programming problem was handled through a process called dualization. Dualization is a technique commonly used in optimization problems to simplify and transform a minimization problem into a maximization problem.

For the first formulation (MXSP-P), which involves maximizing the minimum shortest path length, the initial primal formulation (MXSP-P) was transformed into its dual form (MXSP-D). This was achieved by setting x to a fixed value, deriving the dual of the inner minimization, making some straightforward modifications, and then releasing the fixed value of x. The resulting dual formulation (MXSP-D) is a maximization problem that can be more efficiently solved.

Similarly, for the special case involving the maximization of the minimum detection

probability of traffickers (MXMPD-P), the original non-linear objective function was linearized. The primal formulation was then transformed into its dual form (MXMPD-D) using the same dualization technique.

In both cases, the dualization process simplifies the optimization problems, making them more amenable to solution techniques and allowing for a more efficient handling of the bi-level programming structure. The dual variables introduced in the process play a crucial role in representing and optimizing the objectives of both the upper and lower levels of the bi-level programming models.

3.2. ***Data Collection***

The data collection process for the implementation of the models involved a meticulous study of the Bandhavgarh Tiger Reserve in Madhya Pradesh, India. Leveraging Google MyMap, we accurately identified and mapped the reserve on Google Maps, ensuring a precise representation of the geographical layout. In Figure 1, we can see the map of the Bandhavgarh Tiger Reserve along with the created network for testing the optimization models.

To establish a realistic network for the optimization models, we strategically selected nodes within the tiger reserve. Four sources (s1, s2, s3, s4) were chosen, strategically located deep within the jungle where tigers frequently visit for drinking water and hunting. Additionally, three targets (t1, t2, t3) were identified, positioned near water sources and functioning as transportation hubs with railway stations or as junction points for roads. These target nodes were strategically placed in proximity to larger towns, reflecting potential destinations for traffickers.

The selection of nodes was based on various criteria, including the starting points of roads, intersections of multiple roads within the jungle, and locations with human settlements. This diverse node selection aimed to capture the complexity of the tiger reserve's

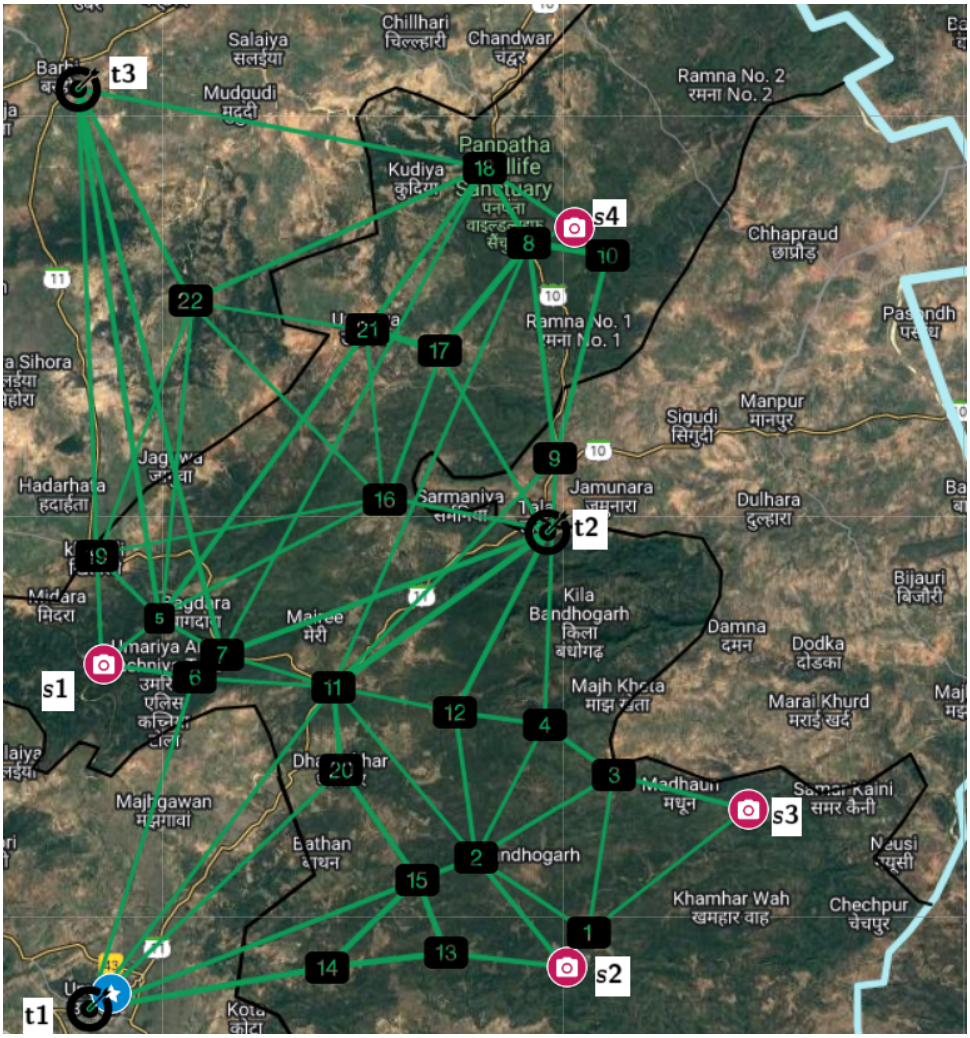


Figure 1: Bandhavgarh Tiger Reserve Map and the synthetically created Network.

landscape and potential trafficking routes. There were 22 additional nodes present in the network. In total, 29 nodes were connected by 68 edges.

For determining the edge lengths (c_{ij}) in the network, we utilized measurements obtained from Google Maps. This approach provided accurate and real-world representations of the distances between nodes, accounting for the twists and turns of the roads within the reserve. An integer delay added to the original length of an edge, denoted as d_{ij}, was set to 500. While a more meticulous selection could be made, we opted for this value.

The assignment of detection probabilities to arcs played a pivotal role in our data

collection process. These probabilities were determined by considering the geographical characteristics of each arc. Arcs situated deep within the jungle were assigned lower detection probabilities, reflecting the challenging terrain and diminished visibility. Conversely, arcs located near larger towns and roads were assigned higher detection probabilities, accounting for the heightened likelihood of human activity and surveillance in those areas. The post-interdiction detection probability (q_{ij}) was deliberately set higher than the pre-interdiction detection probability (p_{ij}). This decision was made thoughtfully, taking into account the specific location of each arc within the jungle. For arcs deep within the jungle, where the terrain poses significant challenges, the increase in detection probability through interdiction is limited. Consequently, the difference between (q_{ij}) and (p_{ij}) is relatively small in these cases. However, for arcs located in other areas with more favorable conditions, the difference is more substantial. This nuanced approach ensures a realistic representation of the impact of interdiction on detection probabilities across diverse geographical contexts.

In conjunction with the careful selection of nodes and determination of edge lengths, the process of selecting interdiction resources was a crucial step in enhancing the realism and applicability of the optimization models. The interdiction resource requirements were assigned as integer values, reflecting the practical constraints associated with real-world interdiction efforts. Our methodology incorporated a strategic approach to resource assignment, directly linked to the length of each edge in the network. Specifically, longer edges, indicative of more remote and challenging terrains within the tiger reserve, were associated with higher resource requirements for interdiction.

This approach aligns with the inherent notion that securing or monitoring longer pathways would demand a greater allocation of resources due to increased logistical challenges and the potential for extended trafficking routes. By introducing integer values for interdiction resources, we aimed to emulate the discrete nature of resource allocation in practical scenarios, where resources are typically finite and allocated in whole units.

This nuanced consideration of interdiction resource assignment not only adds a layer of realism to the optimization models but also acknowledges the varied difficulty levels associated with monitoring or interdicting different arcs within the network. Ultimately, this comprehensive approach to data collection, encompassing node selection, edge length determination, and interdiction resource assignment, ensures that the resulting models are grounded in the unique characteristics of the Bandhavgarh Tiger Reserve, providing a robust foundation for subsequent analyses and decision-making.

3.3. ***Description of the Dataset***

The provided dataset describes a network formulated for modeling the network interdiction problem within the Bandhavgarh Tiger Reserve in Madhya Pradesh, India. The dataset includes sources (s1, s2, s3, s4), targets (t1, t2, t3), and intermediary nodes (1 to 22). Each edge is associated with attributes such as weight (representing the edge length in miles*10), probability of detection without interdiction (pij), probability of detection with interdiction (qij), and the resource required for interdiction (rij). For example, the edge from 's1' to '13' has a weight of 35, a detection probability of 0.4 without interdiction, a detection probability of 0.8 with interdiction, and requires 1 unit of resource for interdiction. All of the values are integers. The dataset comprises 29 nodes and 68 edges.

4. Computational Experiments and Results

4.1. ***maximizing the minimum shortest*** $s-t$ ***path model***

The initial model, solved using the Gurobi solver, focuses on a scenario with zero interdiction budget, effectively reducing the model to a simple shortest path formulation. In this setting, the model produces an optimal value of 91, mirroring the shortest path length provided by NetworkX when no interdiction is in play. This alignment underscores the reliability and precision of our model in capturing and replicating the essential shortest path formulation. The agreement between the Gurobi solution and the NetworkX result at zero interdiction budget serves as a robust validation, confirming the model's accuracy

in representing and optimizing the network's structure for the fundamental shortest path scenario.

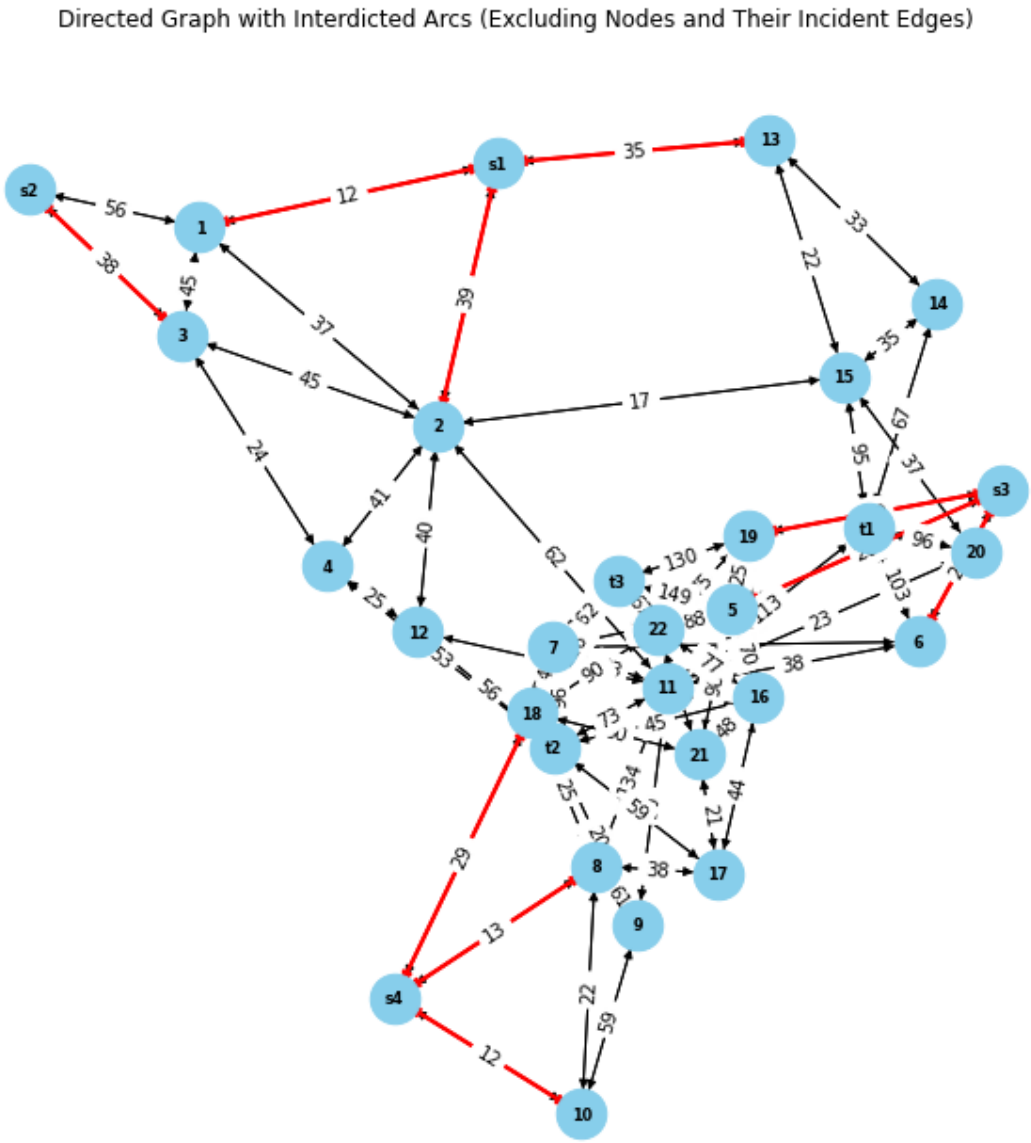


Figure 2: Result of the maximizing the minimum shortest $s-t$ path model with Budget 10

The model was once again solved, this time with a budget of 10. The optimal solution at this budget reveals a strategically interdicted network, impacting specific arcs connecting source and intermediate nodes. Particularly, the arcs (s1, 13), (s1, 1), (s2, 3), (s1, 2), (s3, 6), (s3, 5), (s3, 19), (s4, 8), (s4, 10), and (s4, 18) have been intentionally disrupted. This interdiction pattern is visually represented in Figure 2. The optimal objective value

at a budget of 10 is 178, significantly higher than the shortest path length without interdiction (91). This substantial increase suggests that interdictors should strategically target arcs related to the sources. These arcs are smaller in size, and by interdicting them, the shortest path can be substantially extended. This observation provides valuable insights for decision-makers, emphasizing the critical role of source-related arcs in maximizing the efficiency of the shortest path under specific interdiction scenarios.

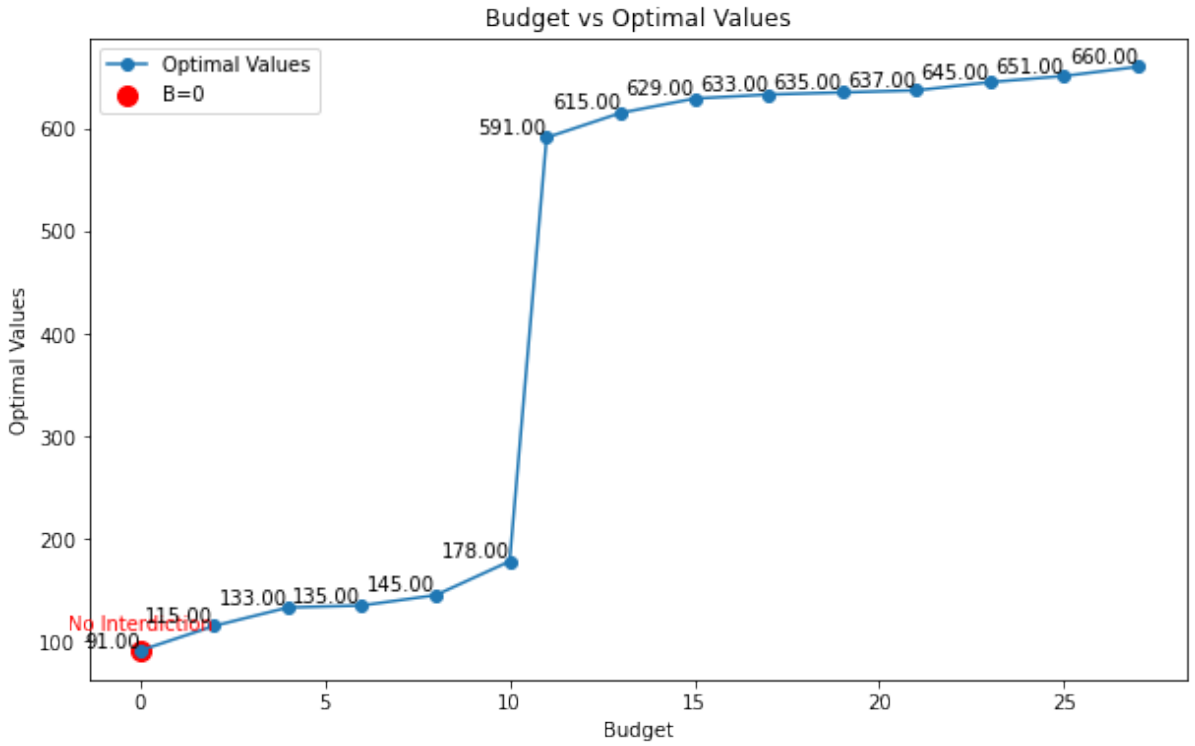


Figure 3: Budget vs shortest path length

In the experiments investigating the impact of the interdiction budget on the shortest path length (Figure 3), a dynamic relationship unfolds. Beginning with a budget of 0, indicating no interdiction, the shortest path length remains at 91.0 units. As the budget incrementally increases, the shortest path gradually elongates until a notable turning point at a budget of 11, where the length sharply rises to 591.0 units. This pivotal shift underscores the pronounced influence of interdiction strategies on the optimal route within the network. This phenomenon can be attributed to the specific setup of the model, where the d_{ij} values for all arcs were set to 500. Consequently, adding 500 to the shortest path length with no interdiction budget (91) results in a cumulative length of

591. This signifies that, as the interdiction budget increases, the trafficker is compelled to traverse through interdicted arcs, ultimately leading to the observed significant increase in path length. Continuing with budget increments, subsequent lengths at budgets 13 and 15 reach 615.0 and 629.0 units, respectively. The marginal changes in path length at higher budgets, such as 17, 19, 21, 23, 25, and 27, indicate a potential saturation point. This suggests that further budget increases may have limited impact on path length.

4.2. ***Special case: Maximizing the minimum detection probability of the traffickers***

In the analysis aimed at maximizing the minimum probability of detection within a budget of 10, specific arcs were strategically interdicted based on notable differences between the probabilities of detection with and without interdiction (q_{ij} and p_{ij}). The selected arcs, including (s2, 3), (s4, 10), (1, 2), (2, 4), (11, t2), (12, 11), (16, 22), and (17, 16), exhibited differences ranging from 0.25 to 0.35. This focused strategy aimed to disrupt critical edges, effectively maximizing the overall impact on the network.

The results, depicted in Figure 4, showed that as the budget increased, the objective values (negative log probabilities) consistently rose, indicating a heightened probability of successful detection. The strategic interdiction approach proved effective in maximizing detection probabilities, with larger negative values representing a higher probability of detection.

The observed trend in the graph, where the curve eventually flattened, aligned with previous findings, suggesting that beyond a certain budget threshold, the incremental impact on the probability of detection became less significant. This underscores the effectiveness of the strategic interdiction approach in balancing budget allocation while maximizing the minimum probability of detection. The approach provides valuable insights for decision-makers seeking to optimize resource allocation for halting wildlife trafficking.

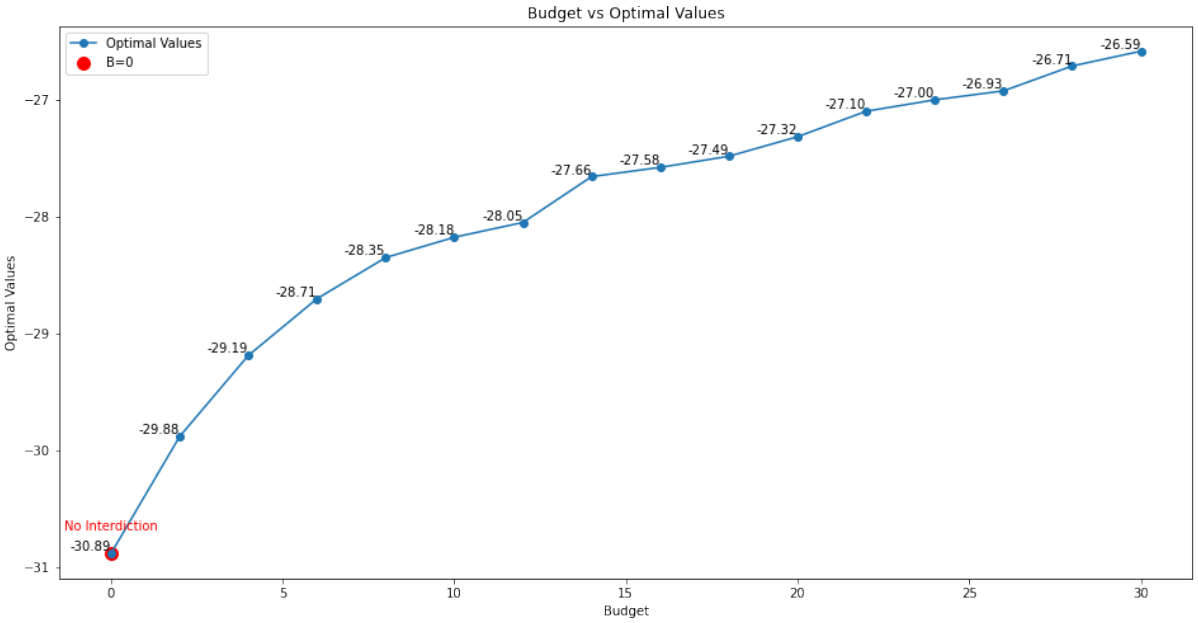


Figure 4: Budget vs sum of the log of dtection probabilities of the shortest path

4.3. ***Time complexity***

The discussion on the time required to solve the models was intentionally omitted due to the rapid solving times of the networks, often taking only fractions of seconds. The negligible time complexity made it unnecessary to delve into detailed discussions on computational efforts for these specific instances.

5. Discussion

Our proposed model addresses a critically important problem – the illicit trade of tiger parts, specifically focusing on disrupting local networks involved in tiger poaching. Tigers, as an iconic species, face the threat of extinction due to wildlife trafficking, and tackling this issue requires a multi-faceted approach. While international efforts have been made to combat the illegal trade, our model hones in on the often overlooked yet vital aspect of local networks that play a significant role in transporting tiger parts from the forest to urban areas.

The results of our model implementation provide valuable insights into the effectiveness of interdiction strategies in disrupting local networks. In the case of the shortest

path formulation, we observed a clear impact of interdiction on the trafficker's optimal route. The strategic disruption of arcs related to source nodes demonstrated a substantial increase in the shortest path length, highlighting the importance of targeting specific nodes for maximum efficacy.

Similarly, in the case of maximizing the minimum probability of detection, our model revealed the effectiveness of a focused interdiction strategy. By strategically interdicting arcs with notable differences in detection probabilities, we maximized the overall impact on the network's security. The results showed a consistent rise in detection probabilities as the interdiction budget increased.

Future directions for this work involve refining and expanding the model to incorporate additional complexities and real-world considerations. One avenue for enhancement is the development of a robust optimization model to address uncertainties in the system. This could account for factors like time windows during which traffickers have limited time to transport poached tigers, introducing a temporal dimension to the model. Moreover, there is potential to enrich the dataset further by including additional relevant information. For instance, incorporating real-time data on tiger movements, detailed terrain characteristics, and other factors influencing detection probabilities would provide a more comprehensive and accurate representation of the Bandhavgarh Tiger Reserve ecosystem.

Moreover, collaboration with wildlife conservation organizations, law enforcement agencies, and environmental researchers could provide valuable insights and data to enhance the model's accuracy and applicability. Integration of advanced technologies such as satellite imagery, machine learning algorithms for predictive analysis, and real-time monitoring tools could contribute to a more dynamic and responsive interdiction model.

Furthermore, the model's adaptability to different geographical locations and species could be explored, addressing broader wildlife trafficking challenges beyond tiger poach-

ing. This could involve adjusting parameters and considerations to suit the specific characteristics of other endangered species and their habitats.

In conclusion, our model offers a targeted and strategic approach to disrupt local networks engaged in tiger poaching, contributing to broader conservation efforts. The results underscore the importance of considering local dynamics in wildlife trafficking and provide a foundation for further research and collaboration in the ongoing fight against illegal trade and the preservation of endangered species.

Appendix A.

Table A.2: Network Dataset for Bandhavgarh Tiger Reserve

Edge	Weight	pij	qij	rij
's1' to '13'	35	0.4	0.8	1
'13' to '14'	33	0.4	0.8	1
'14' to 't1'	67	0.6	0.9	2
'13' to '15'	22	0.5	0.8	1
'15' to '14'	35	0.45	0.8	1
's1' to '1'	12	0.4	0.8	1
's2' to '1'	56	0.3	0.6	1
's2' to '3'	38	0.35	0.7	1
'3' to '1'	45	0.4	0.7	2
's1' to '2'	39	0.3	0.6	1
'1' to '2'	37	0.3	0.6	1
'3' to '2'	45	0.3	0.6	1
'3' to '4'	24	0.25	0.5	1
'2' to '4'	41	0.30	0.55	1
'2' to '12'	40	0.25	0.50	1
'4' to '12'	25	0.25	0.50	2
'4' to 't2'	53	0.30	0.8	2
'12' to 't2'	56	0.25	0.8	2
'11' to 't2'	73	0.35	0.7	2

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Edge	Weight	pij	qij	rij
’2’ to ’11’	62	0.5	0.75	2
’2’ to ’15’	17	0.3	0.70	1
’15’ to ’20’	37	0.5	0.80	1
’20’ to ’11’	23	0.6	0.85	1
’20’ to ’t1’	96	0.6	0.85	2
’11’ to ’t1’	113	0.7	0.95	2
’6’ to ’t1’	103	0.45	0.75	2
’15’ to ’t1’	95	0.45	0.80	2
’12’ to ’11’	35	0.35	0.6	1
’s3’ to ’6’	26	0.40	0.6	1
’s3’ to ’5’	20	0.35	0.6	1
’s3’ to ’19’	30	0.35	0.6	1
’6’ to ’7’	10	0.5	0.6	1
’19’ to ’t3’	130	0.6	0.85	2
’5’ to ’t3’	149	0.4	0.7	2
’7’ to ’t3’	162	0.45	0.8	2
’19’ to ’22’	75	0.4	0.6	2
’5’ to ’22’	88	0.45	0.7	2
’5’ to ’21’	98	0.4	0.65	2
’19’ to ’16’	82	0.4	0.65	2
’5’ to ’16’	70	0.42	0.75	1

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Edge	Weight	pij	qij	rij
’18’ to ’7’	154	0.44	0.65	2
’16’ to ’22’	77	0.35	0.60	2
’21’ to ’16’	48	0.44	0.7	1
’21’ to ’17’	21	0.5	0.75	1
’17’ to ’16’	44	0.4	0.65	1
’17’ to ’t2’	59	0.5	0.75	1
’8’ to ’11’	134	0.45	0.65	3
’7’ to ’t2’	96	0.4	0.68	1
’11’ to ’9’	90	0.65	0.9	2
’9’ to ’t2’	20	0.65	0.9	1
’8’ to ’17’	38	0.4	0.65	1
’8’ to ’9’	61	0.65	0.80	1
’10’ to ’9’	59	0.45	0.65	1
’s4’ to ’8’	13	0.25	0.55	1
’s4’ to ’10’	12	0.25	0.55	1
’s4’ to ’18’	29	0.40	0.60	1
’10’ to ’8’	22	0.25	0.5	1
’18’ to ’21’	60	0.25	0.5	1
’21’ to ’22’	50	0.25	0.5	1
’18’ to ’22’	90	0.30	0.6	2
’18’ to ’t3’	116	0.30	0.6	2

Continued on the next page

Edge	Weight	pij	qij	rij
’22’ to ’t3’	67	0.30	0.5	2
’16’ to ’t2’	45	0.40	0.6	1
’5’ to ’7’	20	0.30	0.55	1
’6’ to ’11’	38	0.5	0.6	1
’7’ to ’11’	33	0.6	0.8	1
’18’ to ’8’	25	0.6	0.8	1
’5’ to ’19’	25	0.5	0.7	1